BUSINESS ANALYTICS CASE STUDY

E-Commerce Customer Churn & Cohort Retention Audit

Analyzing ₹1.10 Crore in gross revenue across 1,200 orders to identify churn drivers and revenue recovery levers.

By Nakshatra Khanna

GROSS MERCHANDISE VALUE (GMV)

₹1.10 Cr

1,200 commercial transactions

60-DAY CUSTOMER CHURN RATE

78.4%

Drop-off after initial purchase

AUDITED CUSTOMER ACCOUNTS

380

Distinct buyer profiles modeled

HIGH-LTV REVENUE AT RISK

₹18.5 L

Trapped in inactive accounts

The Core Business Problem

In digital retail, acquiring new customers costs up to **5× more** than retaining existing buyers. Despite generating healthy top-of-funnel customer traffic, our cohort curves revealed severe downstream revenue leakage.

- **Steep First-to-Second Order Cliff:** Nearly 8 out of 10 first-time buyers never return for a repeat purchase within 60 days.
- **Undifferentiated Marketing:** Generic blanket discount blasts were sent to all users regardless of recency or lifetime spend.

Analytical Scope & Methodology

To resolve this drop-off, I executed an end-to-end data pipeline combining SQL modeling, quantile segmentation, and executive KPI reporting:

SQL Cohort Modeling: Multi-table CTEs & window functions tracking monthly repurchase curves.

RFM Segmentation: Scored accounts 1–5 on Recency, Frequency, and Monetary spend in Python.

The 60-Day Churn Cliff & Drop-off Timeline

Pinpointing when customers disengage between order delivery and permanent churn.

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Order Lifecycle Drop-off Milestones

1. **Day 0: Initial Acquisition & Order Placement**
   100% cohort activation. Average initial order value: ₹916. Customer sentiment is at its absolute peak upon unboxing.
2. **Day 1 – 14: The "Golden Onboarding Window"**
   42% of buyers open post-purchase emails, but only 11.2% view related products. Lack of replenishment triggers.
3. **Day 15 – 24: The Critical Repeat Tipping Point**
   Decisive threshold: 86% of repeat purchases occur before Day 24. If no second order occurs by Day 24, churn probability spikes to 82%.
4. **Day 60+: The Dormant Lapsed State**
   78.4% of first-time shoppers enter inactive state. Reactivation ad cost increases by 4.2× compared to day-10 engagement.

Executive Insights & Key Discoveries

Insight 1: The Day-24 Velocity Cliff
Waiting for day 45 or day 60 to send "We miss you" coupons is too late. The emotional connection expires within 3 weeks of delivery.

Insight 2: Ad Spend Cannibalization
Over ₹4.2 Lakhs was spent retargeting single-purchase dormant buyers with low LTV, delivering an unprofitable 0.6× ROAS.

Insight 3: The 2nd Order Multiplier Effect
Customers who successfully make a **second purchase** demonstrate a **68% retention rate** for orders 3, 4, and 5.

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RFM Segmentation Matrix & Strategy Playbook

Classifying 380 customer accounts by Recency, Frequency, and Monetary value into actionable profiles.

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Customer Segment	% of Base	Revenue Contributed	Behavioral Characteristics	Recommended Growth Action
Champions High Loyalty / High LTV	**32.4%** (123 users)	**₹48.2 Lakhs** (43.8%)	Bought recently, buy often (3+ orders), average order value exceeds ₹1,650.	VIP loyalty tiers, early access to new product drops, priority customer service. No discounts needed.
Promising New Shoppers High Growth Potential	**18.2%** (69 users)	**₹19.4 Lakhs** (17.6%)	Recent 1st order within past 20 days. High engagement but has not repurchased yet.	Trigger automated Day-10 post-purchase drip email with personalized cross-sells & category vouchers.
At-Risk High Spenders Urgent Intervention	**17.6%** (67 users)	**₹28.6 Lakhs** (26.0%)	Historically high spenders who have not placed an order in over 60 days.	Personalized WhatsApp/SMS concierge re-engagement with high-value time-sensitive incentives.
Hibernating / Lapsed Lowest LTV	**31.8%** (121 users)	**₹13.8 Lakhs** (12.6%)	Single discount-driven order, inactive for >90 days. Low email open rates.	Cap paid ad retargeting spend; send only automated quarterly seasonal clearance campaigns.

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Category-Level Churn & Repurchase Correlation

Evaluating repeat purchase rates and average order margins across main product categories.

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Product Category Retention Audit

Category	Repeat Rate %	Avg Order Value	Retention Health
Apparel & Fashion	24.2%	₹1,450	Healthy Anchor
Home & Kitchen	22.8%	₹2,100	High Basket Size
Electronics & Gadgets	18.5%	₹3,250	Long Replacement Cycle
Beauty & Personal Care	16.8%	₹680	High Churn Friction
Books & Stationery	14.1%	₹420	Single-Item Gifts

Strategic Merchandising Discoveries

1. Apparel Drives Organic Re-engagement

Customers purchasing apparel return 2.4× faster than other categories. Apparel serves as the ideal gateway category for subscription and loyalty clubs.

2. Beauty Requires Auto-Replenishment

Beauty items have high consumable turnover (30–45 day lifecycles), yet suffered an 83.2% churn rate due to zero replenishment reminders.

3. Cross-Category Bundling Opportunity

Buyers who ordered from **two or more categories** exhibited a **52.4% repeat purchase rate**, compared to just 18.2% for single-category shoppers.

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Strategic Recommendations & Expected Impact

Three data-backed growth initiatives designed to recover revenue and boost 60-day customer retention.

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Initiative 1

Day-10 Cross-Sell Drip Workflow

Deploy automated post-delivery email/SMS sequences triggered exactly on Day 10 offering tailored recommendations based on initial order category.

PROJECTED OUTCOME

+12.4% Repurchase Rate

Captures buyers before the Day 24 cliff.

Initiative 2

Reallocate Ad Budget to Retention

Shift 20% of top-of-funnel acquisition ad budget into targeted re-engagement campaigns focusing exclusively on the 67 "At-Risk High Spenders".

PROJECTED OUTCOME

₹6.8 Lakhs LTV Recovered

Protects high-value customer accounts.

Initiative 3

Gamified 30-Day Velocity Points

Introduce a "Fast-Track Loyalty Reward": Award 2× bonus reward credits for any repeat purchase placed within 30 days of the previous order.

PROJECTED OUTCOME

-18 Days Purchase Cycle

Accelerates repeat buying frequency.

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